

AI5707 – Pattern Recognition and Machine Learning

Assignment 5: Maximum likelihood parameter estimation

Due: Oct 05 2026

1 Objective

In this assignment, you are expected to perform parameter estimation on a) the data generated in the last assignment using the method of maximum likelihood, b) some real data.

2 Problem definition

1.
 - (a) Generate about 10000 points using the program from the last assignment for each of three different classes $\mathcal{N}(\mu_i, \Sigma_i), \forall i = 1, 2, 3$
 - (b) Use different types of covariance matrices, namely (a) $\sigma^2 \mathbf{I}$, (b) $\Sigma_{ii} = \sigma_{ii}^2$ on the diagonal, all other values being zero. (c) Full covariance matrix.
 - (c) Generate data with identical covariance matrices across classes, and different covariance matrices across different classes.
 - (d) Divide the data into 80% train, 20% test. Estimate parameters using train data.
 - (e) Using the models generated in the previous step, classify the 20% test data.
 - (f) Generate a plot with boundaries between classes, and overlay the test data on this plot.
 - (g) Generate a table corresponding to the confusion matrix.
 - (h) Record your observations.
2. Use the 2-D data available at the link:
 - (a) <https://drive.google.com/drive/folders/115yf7pczFx0D4M6-SLaHWjAyEIDob7bN>
The filename suffix should match your team number.
 - (b) The data is given in the format x,y, class label. Estimate parameters using maximum likelihood estimation using 80% of the data.
 - (c) Classify the remaining 20%. Generate the confusion matrix. Plot the boundaries between the classes. Overlay test data on this plot.
 - (d) Record your observations.